Instructor Key: Day 3 Mastery Quiz

Types, Casting, and Truthiness

Key for the Day 3 student quiz package.

1 Artifact status and source boundary

Field	Status
Student artifact	10-point Day 3 mastery quiz; quiz, not workbooklet.
Content anchor	Built after Lance approved the Day 3 workbooklet content surface: types, casts, truthiness, non-mutating type checks, and safe evidence before threshold decisions.
Difficulty posture	Higher than workbooklet recall: uses transfer values, string concatenation trap, non-mutating cast distinction, and non-empty string truthiness trap.
Scenario/source boundary	Lower-sensitivity pump-station intake record. No protected exam/review prompt text, answer-key language, private student data, or near-copy source structure used.
Student-surface rule	Student PDF intentionally hides internal file versioning and release/validation language; visible student surface carries Lance copyright.
Canvas status	Not Canvas-released. Student PDF still needs Lance upload/staging plus Ally and Panorama review if selected for use.
Copyright	Copyright © 2026 Lance L.A. White. All rights reserved.

2 Reference code

```python
# intake values from the station form
pump_id = "P-31"
target_text = "21.0"
reading_text = "19.8"
tolerance_text = "0.6"
approval_text = "False"
comment_text = ""
retry_count = 0
sensor_ready = True

# conversions and checks for the next day's limits
target_lpm = float(target_text)
reading_lpm = float(reading_text)
tolerance_lpm = float(tolerance_text)
gap_lpm = target_lpm - reading_lpm
approval_flag = bool(approval_text)
comment_flag = bool(comment_text)
retry_flag = bool(retry_count)
```

3 Point map, secure answers, and support moves

Pt.	Item target	Secure answer	Common issue / support move	Tags
1	Type before conversion	`reading_text` is a string / `str`. It stores `"19.8"`, not numeric `19.8`.	If student says float, ask whether Python has run `float(...)` on that variable yet.	recognize
1	Cast result	`target_lpm` stores numeric `21.0`; type is `float`.	If student writes string, point to the assignment target on the cast line.	trace; predict
1	Non-mutating cast expression	`float(tolerance_text)`. This produces `0.6` without changing `tolerance_text`.	If student includes assignment, accept only if they explicitly distinguish expression from storage under local grading.	implement; trace
1	Original text after cast	`target_text` still stores `"21.0"`. Casting elsewhere did not rewrite the source text variable.	Key Day 3 misconception: conversion result changes the original string automatically.	trace; explain
1	Numeric gap	`gap_lpm = 1.2`. Calculation: `21.0 - 19.8`.	Allow small floating-representation awareness only if clear; target answer is `1.2`.	predict
1	String concatenation trap	`target_text + tolerance_text` produces `"21.00.6"`. It concatenates strings rather than adding numeric values, so it is not upper limit `21.6`.	This is the intended higher-difficulty trap. Require both produced value and reason.	debug; explain
1	Non-empty string truthiness	`approval_flag` stores `True`, because `"False"` is a non-empty string. It is not approval evidence.	If student says false, ask whether Python reads the word meaning or only empty/non-empty status for truthiness.	predict; debug
1	Falsey empty/zero values	`comment_flag = False`; `retry_flag = False`. Both come from falsey values: empty string and numeric zero.	If student treats `0` as text, revisit type before truthiness.	predict
1	Unsafe note	Mistake 1: `approval_flag` is truthiness of non-empty text, not approval. Mistake 2: `target_text + tolerance_text` concatenates strings; numeric upper limit requires converted values, e.g., `target_lpm + tolerance_lpm`.	Require both mistakes. One-mistake answers are developing, not secure.	debug; test
1	Explicit approval check	Accept `approval_text == "True"`, `approval_text == "approved"` if paired with a defined label convention, or another exact-label comparison. In this code, `approval_text == "True"` evaluates `False`.	If student writes `bool(approval_text)`, circle the non-empty string trap.	implement; transfer

4 Derived values and truthiness trace

Name/expression	Value	Type/result	Reason
`float(target_text)`	`21.0`	float	Cast produces numeric value from text.
`float(tolerance_text)`	`0.6`	float	Cast produces numeric tolerance.
`target_text + tolerance_text`	`"21.00.6"`	string	String concatenation.
`bool(approval_text)`	`True`	bool	Non-empty strings are truthy.
`bool(comment_text)`	`False`	bool	Empty strings are falsey.
`bool(retry_count)`	`False`	bool	Numeric zero is falsey.

5 Mastery interpretation and support routing

Band	Point guide	Evidence pattern	Next support action
Secure	8–10 / 10, including both debug traps	Student can distinguish text from numeric casts, explain non-mutating conversion, and reject truthiness as approval evidence.	Continue to Day 4 boundary checks with converted numeric fields.
Developing	5–7 / 10, or one debug trap missed	Student can name some types but may confuse text meanings with Boolean meaning or string concatenation with numeric addition.	Target type/cast/truthiness; run three one-line checks: `type(...)`, `float(...)`, `bool(...)`.
Needs support	0–4 / 10	Student does not yet separate source text fields, converted numeric fields, and truthiness results.	Model a two-variable text-to-float example and an empty/non-empty string truthiness example.

6 Optional recovery mini-surface

Use if several students treat text meaning as Boolean meaning.

```
label = "no"
blank = ""
number_text = "0"
count = 0
bool(label)       # True
bool(blank)       # False
bool(number_text) # True
bool(count)       # False
```

Secure recovery answer: truthiness does not read the English meaning of a string. It checks empty versus non-empty for strings, and zero versus nonzero for numbers.

7 Release/accessibility notes

Local verification should check tagged PDF output, extractable text, page count, print-safe margins, render sheets, answer-key separation, no private/protected markers, and the student-facing no-internal-surface rule. If selected for Canvas use, Lance uploads/stages the student PDF and retrieves both Ally and Panorama scores. Working release threshold remains Ally at least 95% and Panorama at least 95%.